Improving Infectious Disease Prevalence Estimation and Parameter Inference Using Number of Tests and Positivity Data

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Overview

- Introduction
  - Prevalence
  - Non-random testing
- Top-Down Approach
- Bottom-Up Approach
  - Hazard model
  - Odds ratio model
  - Fitting showcases

Prevalence

- Prevalence $Y(\mathrm{t})$: Proportion of infectious individuals in the population
  - Dynamic modeling: Determines the force of infection
  - Data fitting: Works as prediction while data as observation
  - Decision-making: Important status indicator of the outbreak
- Challenging to measure or deduce Y from the data of large population:
  - Only able to observe cases being tested and reported
  - Little information about the untested population
- Common naive approaches:
  - Assume $Y =$ test positivity P i.e. proportion of positives among all tested cases
    - Implicitly indicates that the tests are distributed uniformly random to the whole population
  - Assume all cases are reported or being reported at a fixed portion
    - Small/controlled population or disease with obvious symptoms

Non-random Testing

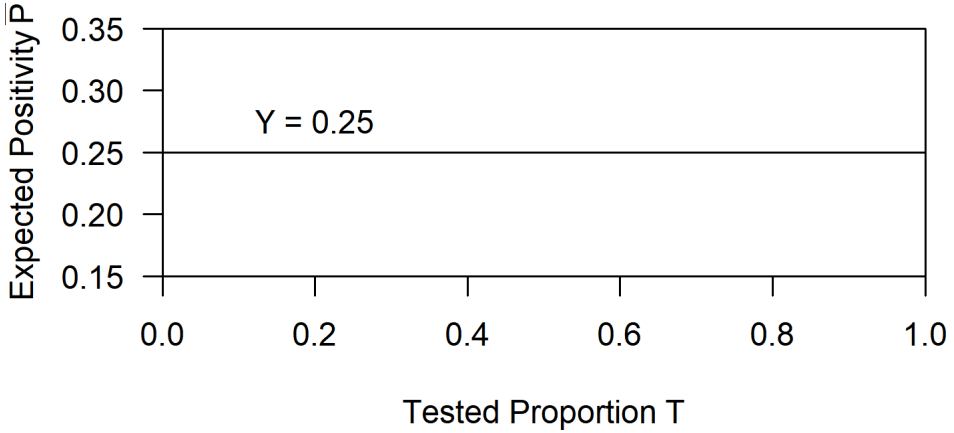
- Many infectious disease tests are not distributed at random
  - How tests are distributed could affect prevalence and positivity
- Symptoms:
  - Symptom severity↑ → the more likely someone is to seek medical care and tests
  - Sub-clinical cases might be common in infected cohort, but rarely being tested
- How cases got infected:
  - E.g. for high-profile infection, imported cases and close contacts of existing cases are more likely to be tested
  - An asymptomatic, community-infected person might never be tested
- **Why and how people are tested** is important when estimating prevalence with non-random testing

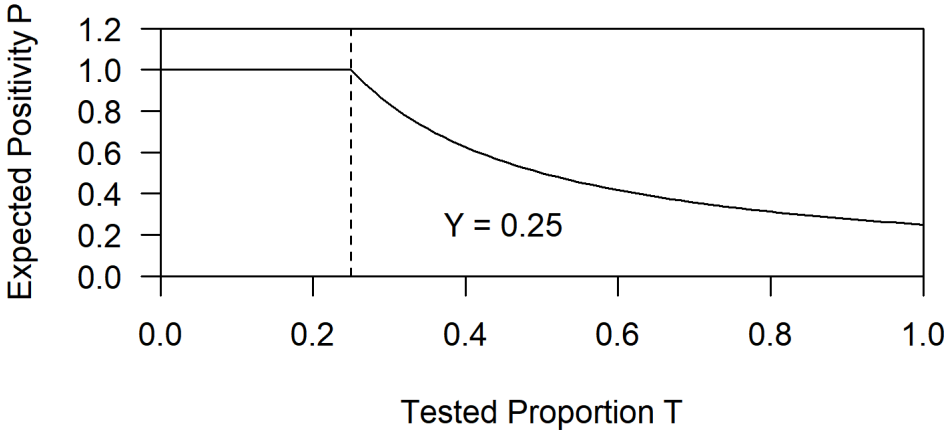
Testing Mechanism Models

- Many related factors:
  - Delay from infection event, to becoming infectious and to being tested
  - Heterogeneity in testing behavior and test policies
  - Impact of public awareness and medical resources
- Start with building basic, simple models
  - Focus on testing mechanics and priority
  - Use simple models characterized by on 1-2 parameters
- For each reporting period of data, connect prevalence Y with:
  - T: Proportion of tested individuals among whole population
  - P: The test positivity
  - T and P can be deduced from data of the number of conducted tests and positive tests

Top-Down Approach

- Consider how tests are distributed:
  - More proper for the situation that tests are limited
  - Phenomenological approach
- Two extreme scenarios:
  - **Totally random testing**: the expected positivity $\overline{\mathrm{P}}$ = the prevalence Y
  - **Perfectly focused testing**: test is perfectly distributed to infected people first
    - Tests are 100% positive until we run out of infected people (i.e. $T > Y$)
    - The expected positivity is $\overline{\mathrm{P}} = \min(1, Y/T)$





Beta-Distribution Model

- Assume people are tested in order of their probability of being infectious
  - Ignoring tests skipped because of constraints
- We can think about:
  - A distribution for the probability of infection in the population
    - E.g. higher for known contacts of infected people, travelers from high-risk areas, etc.
  - A distribution in the probability of testing
    - Ideally lines up with the risk
- We could use a single beta-distribution to combine them, with PDF:

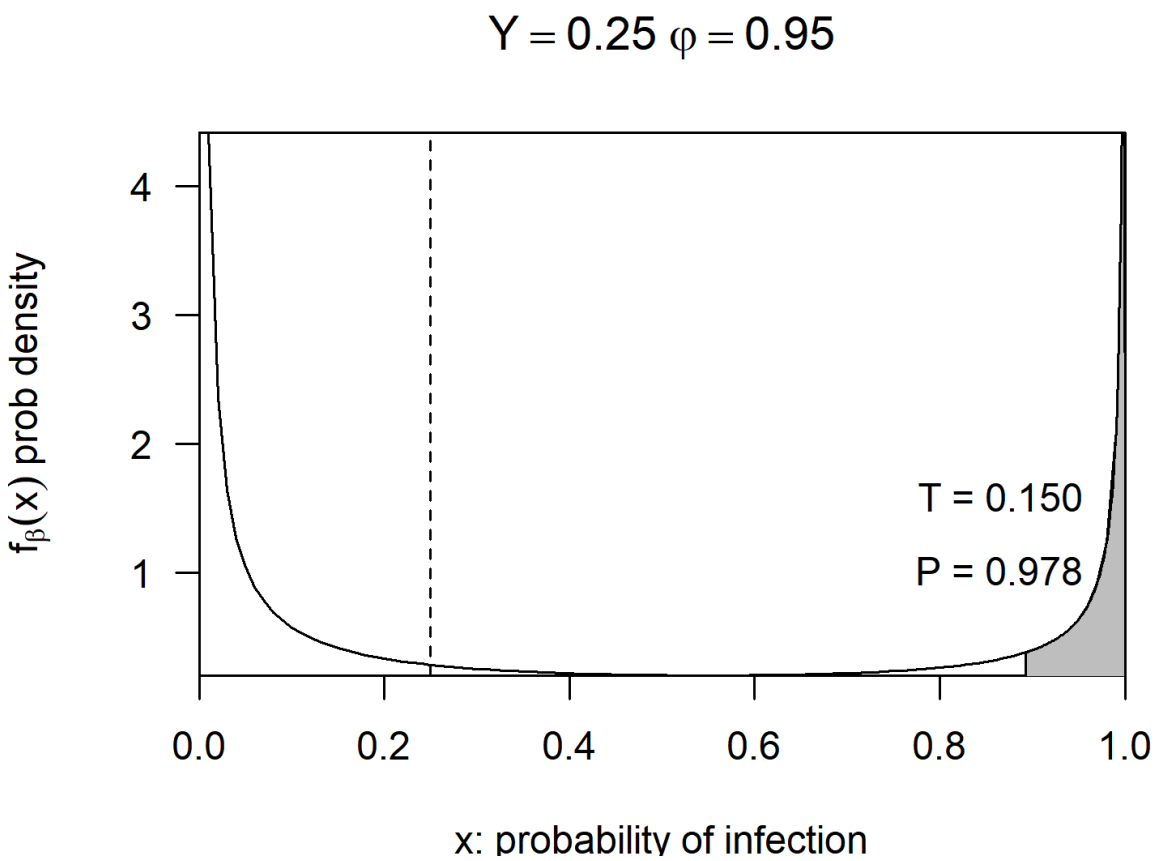
$$f_\beta(x; a, b) \propto x^{a-1}(1-x)^{b-1}$$

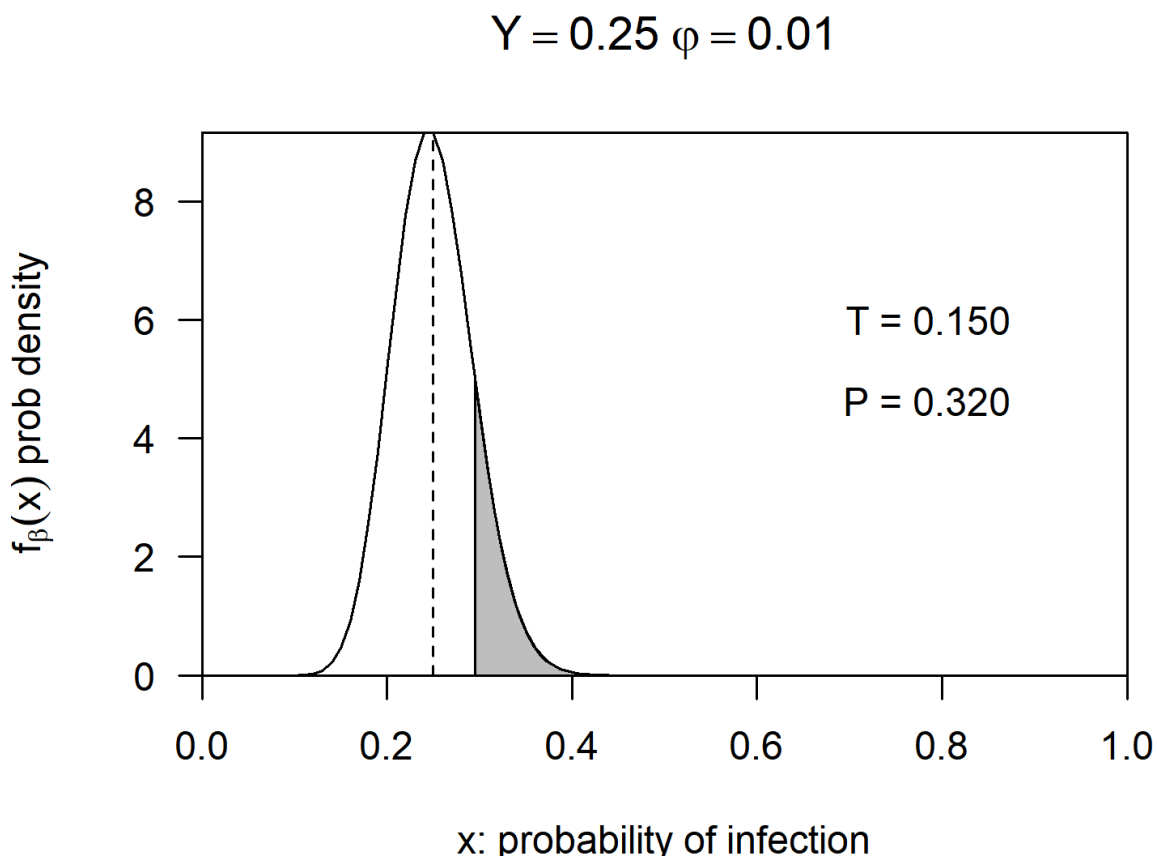
  - $x \in [0,1]$, higher x means higher risk $\Leftrightarrow$ higher probability of being infectious
  - f_β gives the population density with probability x of being infected

Parameterization with Testing Process

- For our model, $f_\beta(x; a, b)$ could be parameterized by:
  - Its mean $E[X] = \frac{a}{a+b} = Y$ equals to the prevalence
  - Its dispersion $\varphi = 1 - e^{-\frac{1}{a+b}}$ characterizes the "testing focus“
- If we always test 'from the top down’,
  - Positivity is the expectation of the top T fraction
  - As $\varphi \to 1$ we end up with point masses $1 - Y$ on $x = 0$ and Y on $x = 1$, correspond to perfectly focused testing ($PT = min(T, Y)$)
  - As $\varphi \to 0$ we end up with a point mass on Y, correspond to totally random testing ($P = Y$)

Impact of Testing Focus φ





- More focused test (larger ϕ) leads to higher expected positivity $\bar{P}$ when testing the same proportion T

Expected Testing Positivity $\bar{P}$

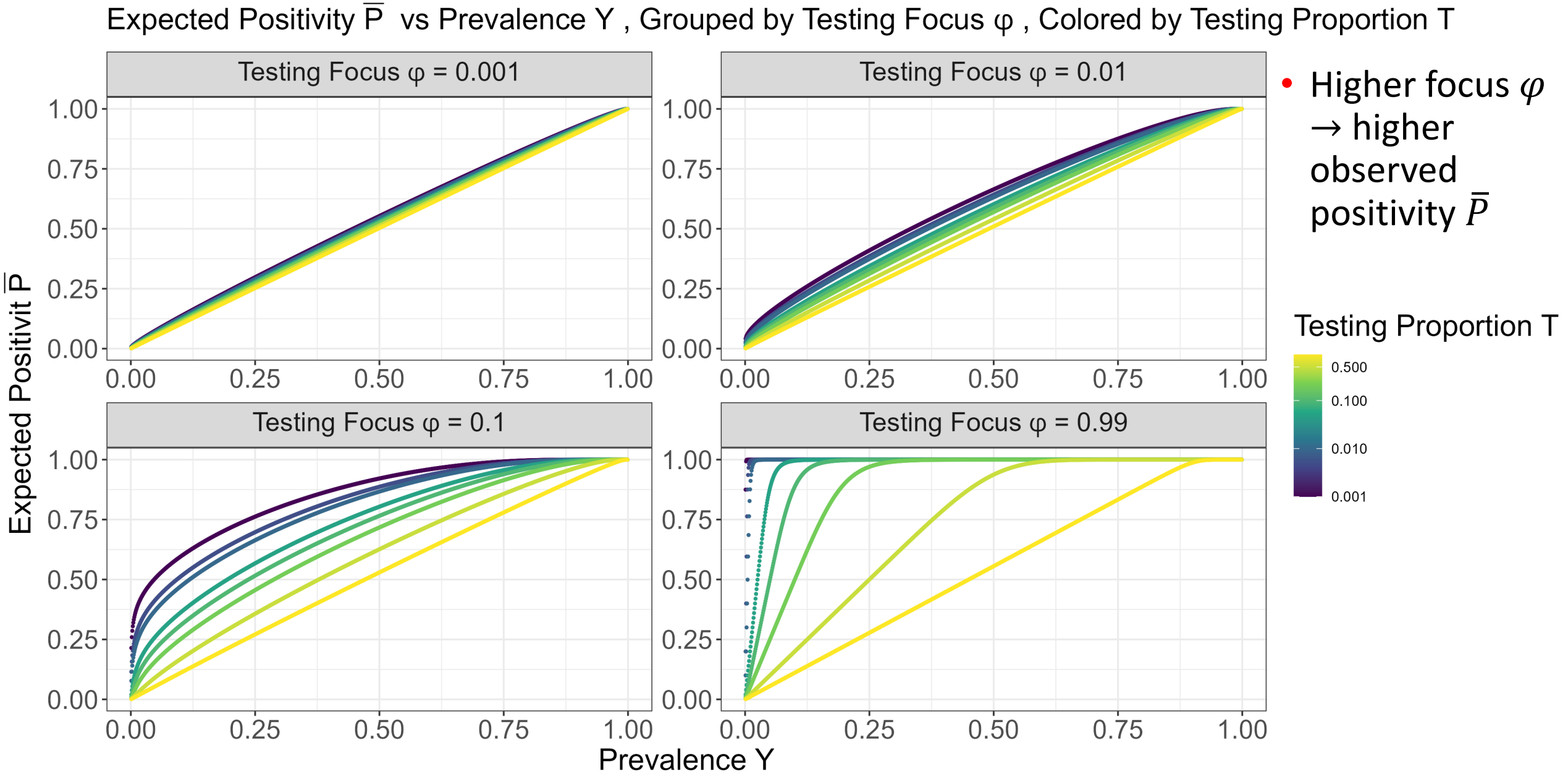
- The expected testing positivity $\bar{P}$ is the mean density of the tested portion (grey area):

$$\bar{P} = \frac{\left(\int_{Q_\beta(1-T)}^{1} f_\beta(x; Y, \varphi) x \; dx\right)}{T} = Y\left(1 + \frac{Q_\beta(1-T)^a \left(1 - Q_\beta(1-T)\right)^b}{Ta\,\mathcal{B}(a,b)}\right)$$

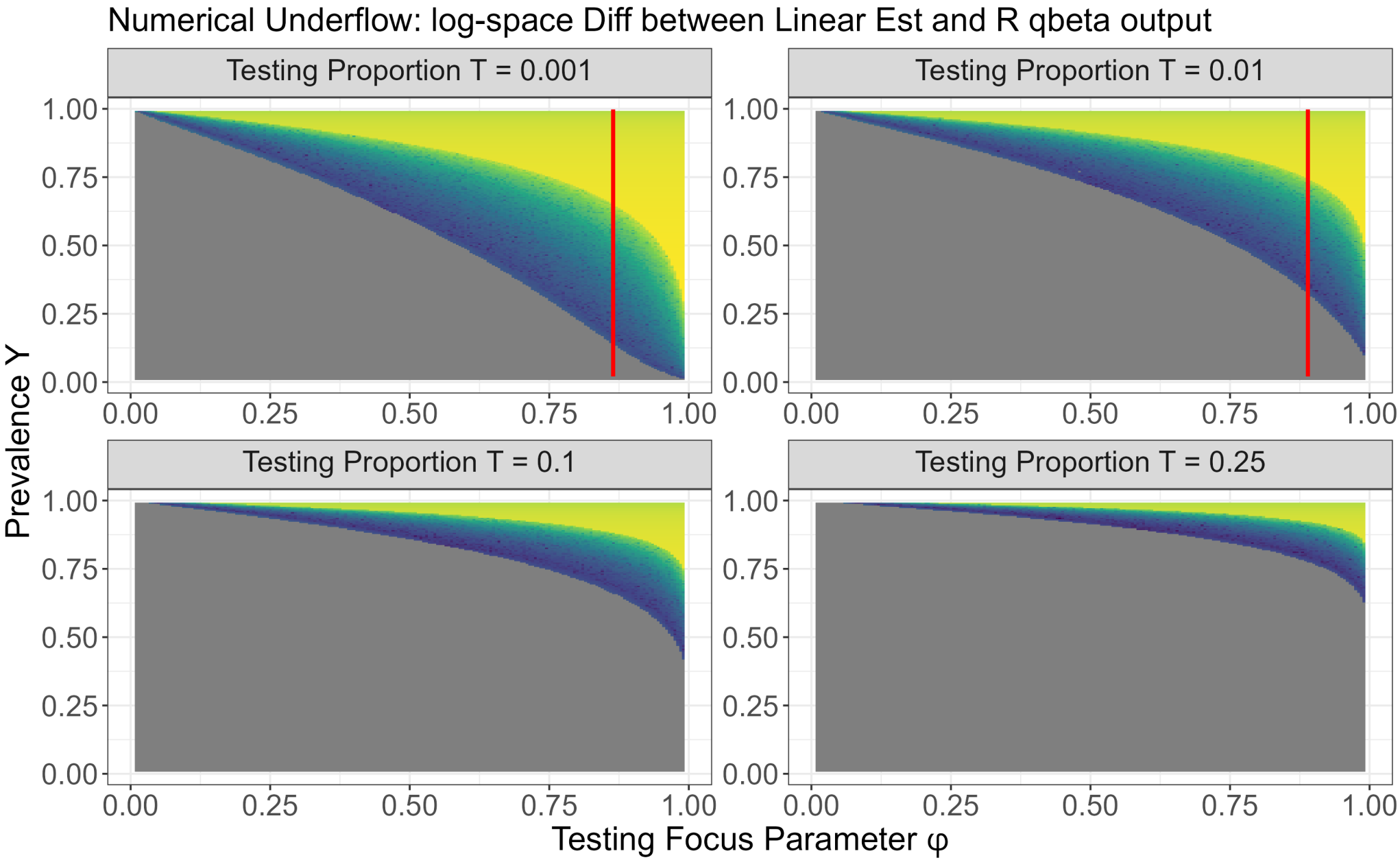
  - F_β is the corresponding CDF
  - $Q_\beta(z)$ is the z-quantile of the distribution
  - $\mathcal{B}(a,b)$ is the complete beta function

- As $Q_\beta \to 0$, qbeta function would have numerical underflow to 0 due to floating-point, we could use a first order approximation:

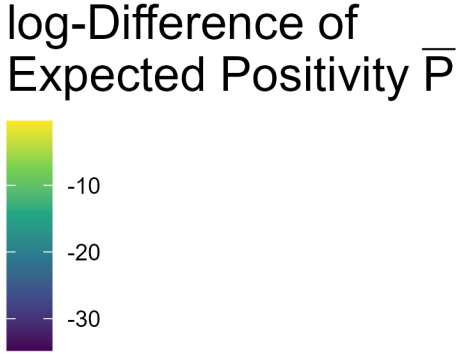
$$\bar{P} \approx \frac{b + 1 - ab}{b + 1 - (1-a)b}$$



- Higher focus φ → higher observed positivity $\bar{P}$



- Grey: qbeta output is numerically consistent



- Blue: occasional underflow
- Yellow: significant underflow

Pros and Cons

- (+) Testing process is characterized by one parameter φ
- (+) Simple and interpretable mechanism with flexibility
- (−) Highly rely on the assumption for testing priority
  - Might only be proper for limited number of tests
- (−) Hard to interpret, measure and change φ in real-world
  - φ can only be achieved by fitting
  - Hard to modify φ for testing policy shifts
  - Not ideal for fitting a varying φ
- (−) Performance concern for fitting data
  - Numerical underflow issue
  - Change of prevalence Y will also change the distribution

Bottom-Up Approach

- Model the probability of people seeking care for various reasons
  - More proper for the situation that tests are unlimited and more voluntary
  - Mechanistic approach
- Consider two complementary cohorts of the population
  - Uninfected population with a probability T_{B} of seeking tests and care
  - Infected population with a probability T_{Y} of seeking tests and care
  - In general, $T_{\mathrm{Y}} \geq T_{\mathrm{B}}$
- Assuming each cohort is homogeneous, then consider prevalence Y
  - The total testing proportion/probability is

$$T = T_{\mathrm{B}}(1 - Y) + T_Y Y$$

  - The expected test positivity would be

$$\bar{P} = \frac{Y T_Y}{T}$$

Hazard

- Hazard is the probability that an event will occur at a specific time, given that it has not occurred yet
  - For an event with probability $\mathcal{P}$, its hazard is given by

$$H = -\log(1 - \mathcal{P}) \Leftrightarrow \mathcal{P} = 1 - \mathrm{e}^{-H}$$

  - For an event with multiple hazard components:

$$H_{event} = \sum_{c} H_c$$

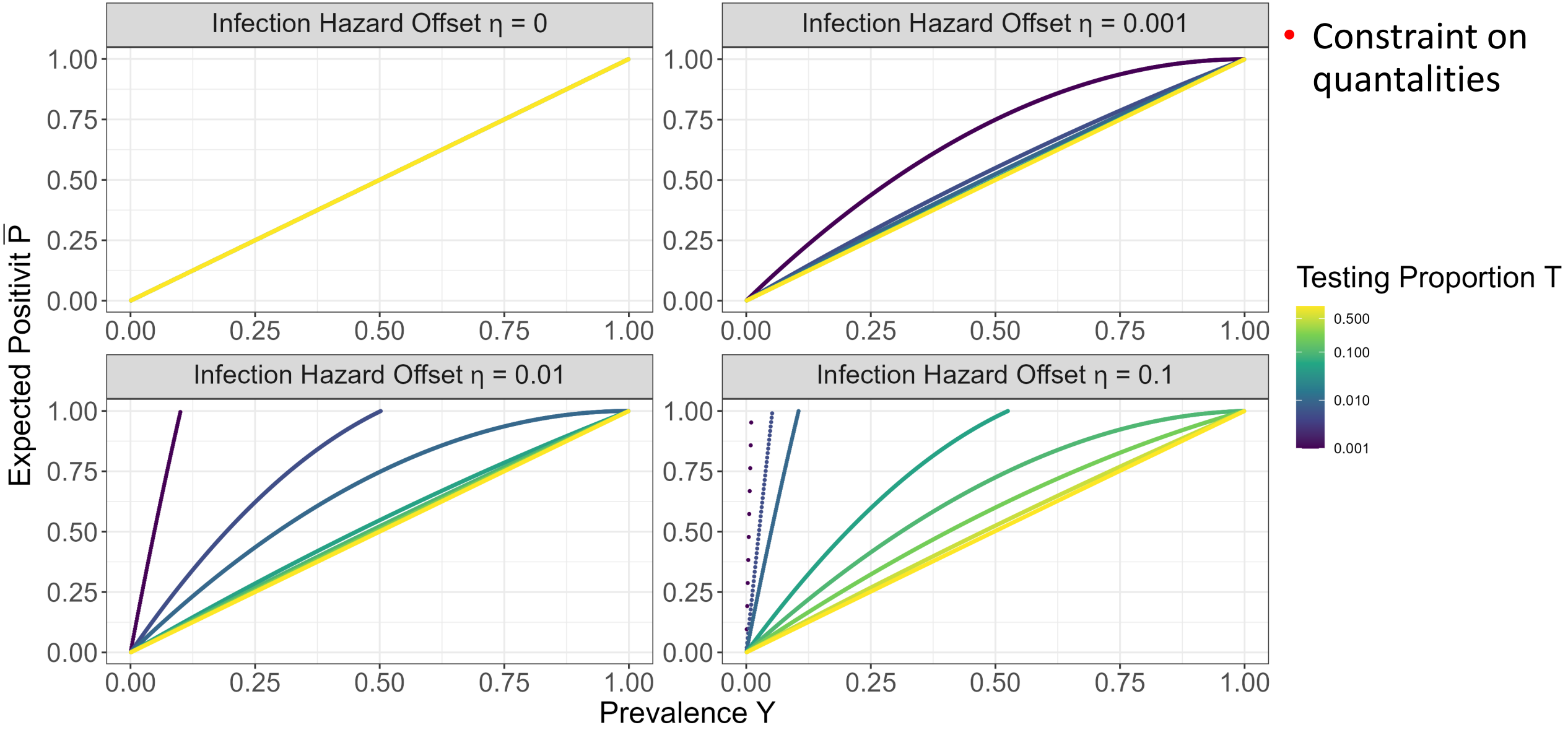
    - E.g., $H_{\text{missingTalk}} = H_{\text{forgetting}} + H_{\text{401traffic}} + H_{\text{gettingLost}}$

$$\mathcal{P}_{\text{missingTalk}} = 1 - (1 - \mathcal{P}_{\text{forgetting}})(1 - \mathcal{P}_{\text{401traffic}})(1 - \mathcal{P}_{\text{gettingLost}})$$

Hazard Model

- For the bottom-up approach
  - Baseline testing probability T_B is governed by the baseline hazard $\mathrm{b} > 0$:
$$T_B = 1 - e^{-b}$$
  - Infected testing probability T_Y is governed by b and **a const. risk offset $\boldsymbol{\eta} > \mathbf{0}$** caused by infection:
$$T_Y = 1 - e^{-(b+\eta)}$$
    - For simplicity, let $B = e^{-b}$ and $\Phi = e^{-\eta}$
  - As a result, we have
$$T = T_B(1 - Y) + T_Y Y = (1 - Y)(1 - B) + Y(1 - B\Phi)$$
  - Then the expected positivity is given by:
$$\overline{\mathrm{P}} = \frac{YT_Y}{T} = \frac{Y(1 - B\Phi)}{T} = \frac{Y}{T}\left(1 - \frac{(1 - T)\Phi}{1 - Y + Y\Phi}\right)$$

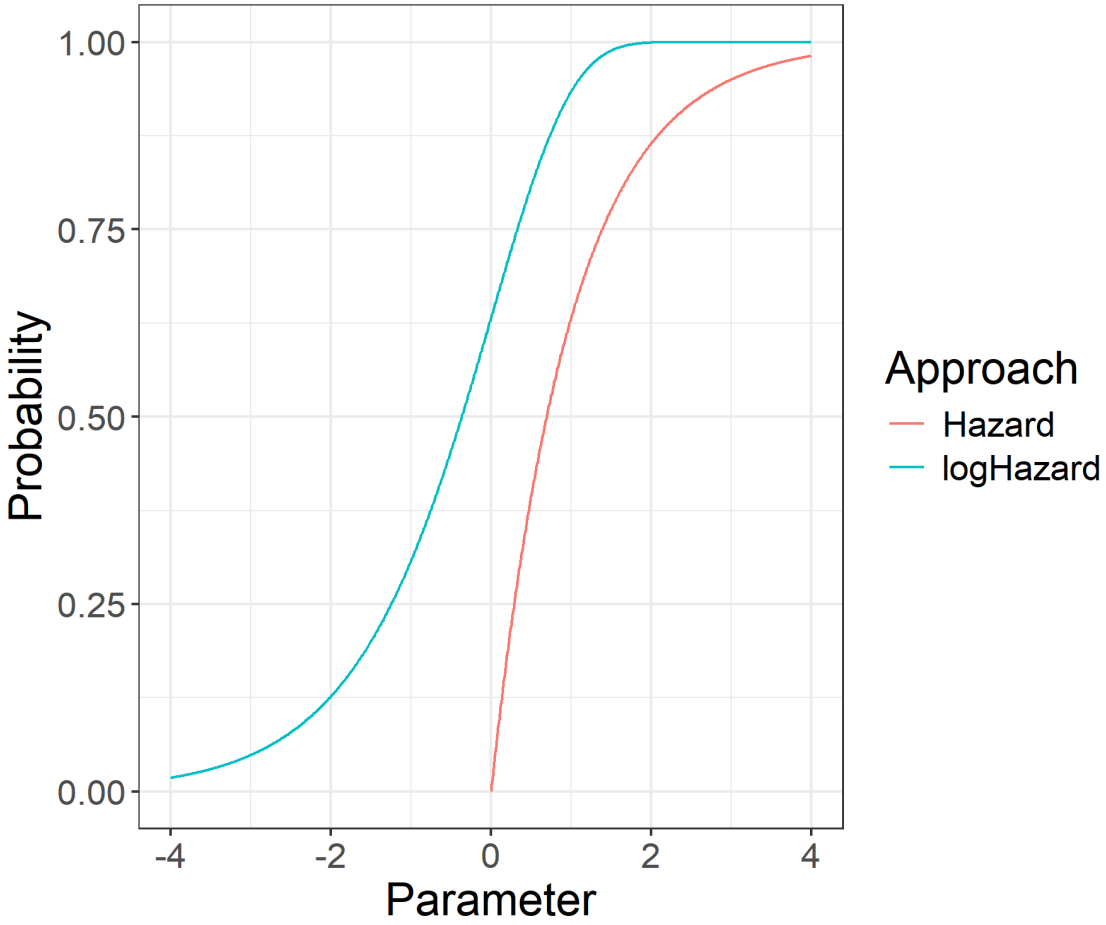
Expected Positivity $\overline{P}$ vs Prevalence Y , Grouped by Infection Hazard Offset η , Colored by Testing Proportion T



- Constraint on quantalities

Pros and Cons

- Hazard
  - (+) Usage is relatively easy to justify
  - (+) Simple expression
  - (−) Probability curves not ideal for fitting
    - Steep drops near 0
    - “suddenly terminate” at 0
- Log Hazard
  - (+) Avoid the “0” problem of hazard
  - (−) Unfavorable behavior in positive interval
    - $\mathcal{P} \to 1$ fast near 2 and flat thereafter
  - (−) Harder to justify usage



Odds

- Odds provide a measure of the probability of a particular outcome
  - For an event with probability $\mathcal{P}$, its odds is given by

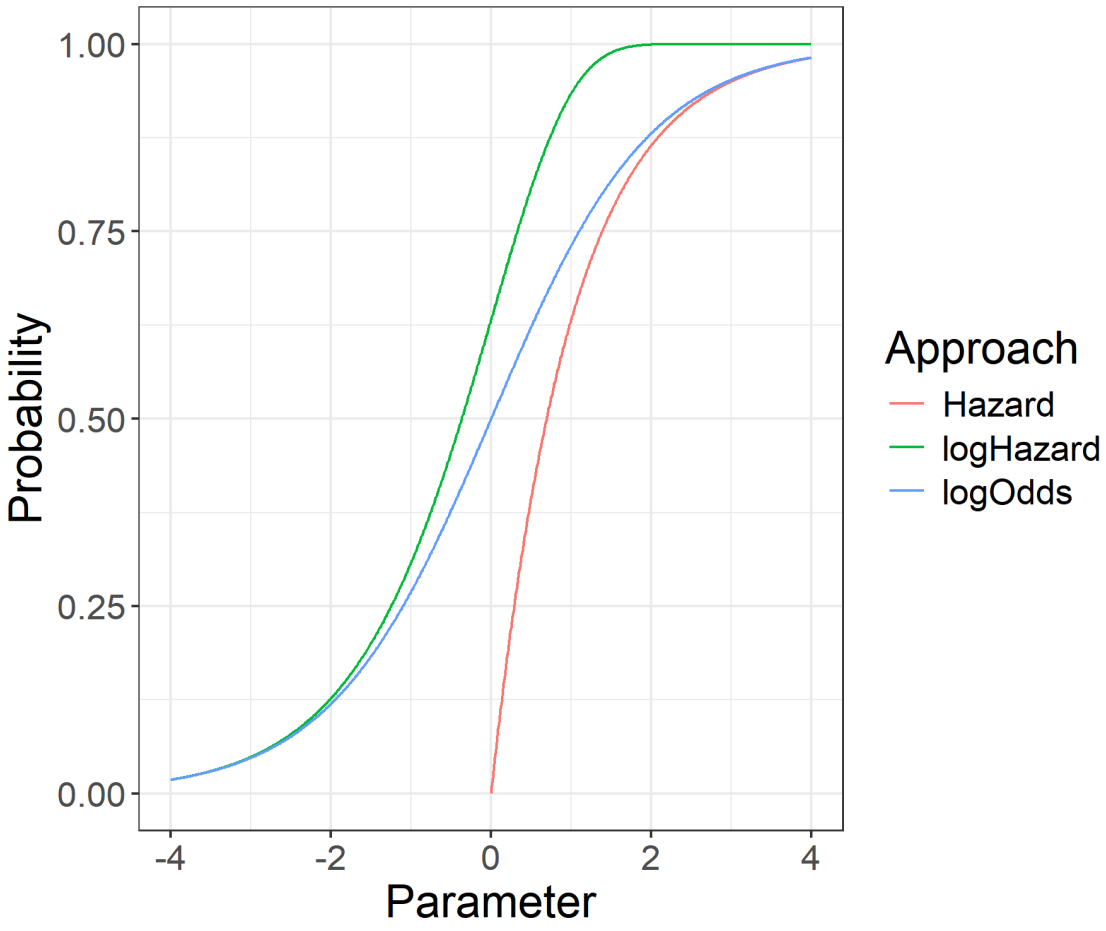
$$\omega = \frac{\mathcal{P}}{1-\mathcal{P}}$$

    - Log odds $\ell = \log(\omega) = \log\left(\frac{\mathcal{P}}{1-\mathcal{P}}\right)$
  - The odds of uninfected individual being tested is

$$B = \frac{T_B}{1-T_B} \Leftrightarrow T_B = \frac{B}{1+B}$$

  - Similarly, the odds of infected individual being tested is

$$\omega_{\mathrm{Y}} = \frac{T_{\mathrm{Y}}}{1-T_{\mathrm{Y}}}$$



Odds Ratio Model

- Odds ratio is a statistic that quantifies the strength of association between an infection exposure and an event
  - For our case, the event is an individual being tested
  - Assume there is **a const. odds ratio** Φ between infected and uninfected individuals, s.t.

$$\Phi = \frac{\omega_Y}{B} = \frac{\frac{T_Y}{1 - T_Y}}{\frac{T_B}{1 - T_B}} \Leftrightarrow T_Y = \frac{B\Phi}{1 + B\Phi}$$

    - $\Phi \geq 1$
    - $\lambda = \log(\Phi) = \log(\omega_Y) - \log(B)$ is also constant

Odds Ratio Model

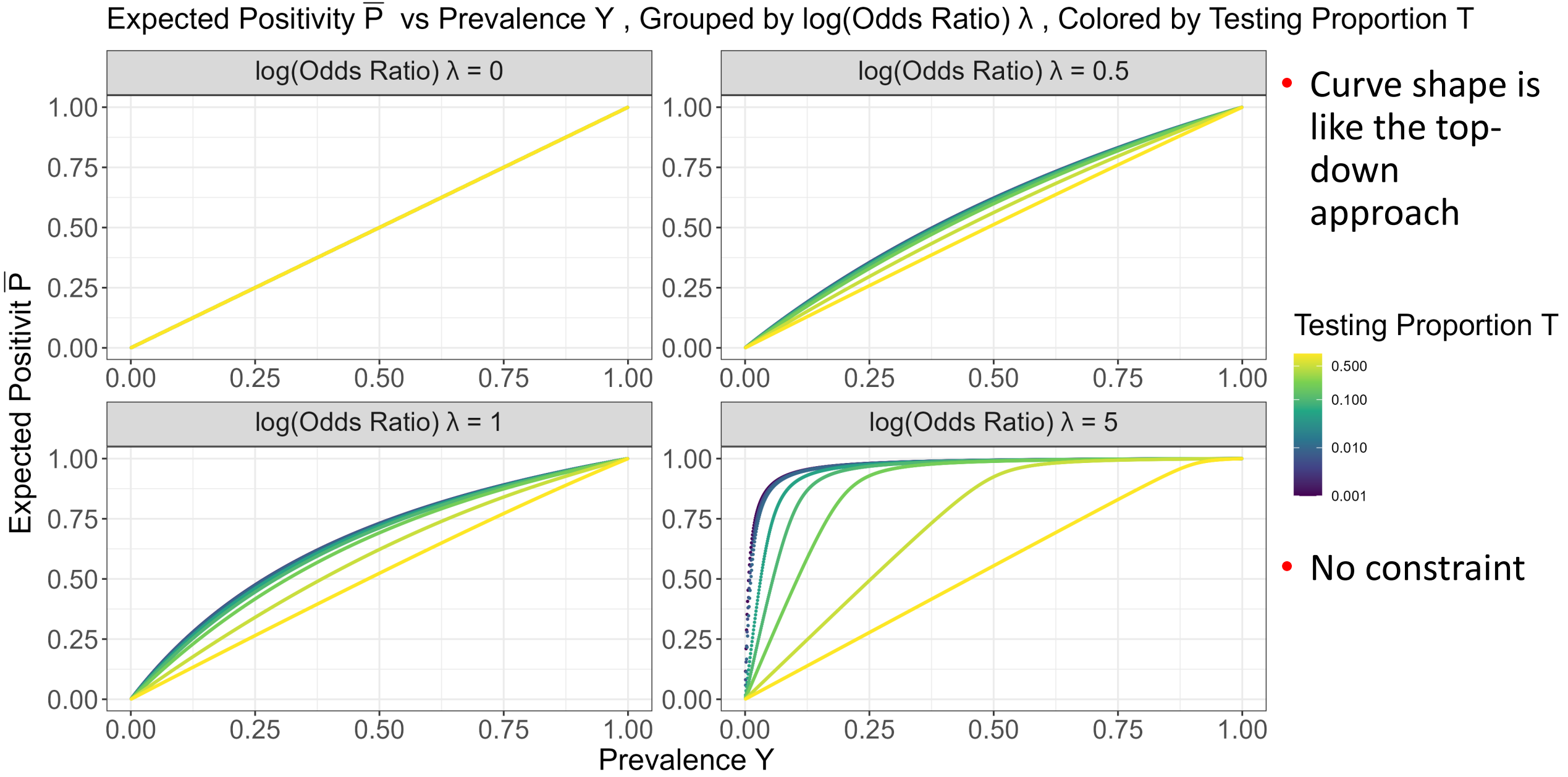
- Then for the testing proportion of the whole population

$$T = (1-Y)T_B + YT_Y = (1-Y)\frac{B}{1+B} + Y\frac{B\Phi}{1+B\Phi}$$

  - Then we can deduce the expected testing positivity

$$\bar{P} = \frac{YT_Y}{T} = \frac{Y}{T}\frac{B\Phi}{1+B\Phi} = \frac{Y}{T}\frac{X+\sqrt{X^2+4T(1-T)\Phi}}{X+\sqrt{X^2+4T(1-T)\Phi}+2(1-T)}$$

    - $X = (T-Y)\Phi + T + Y - 1$



- Curve shape is like the top-down approach
- No constraint

Fitting Bottom-up Model

- In most testing data sets, we are only able to observe
  - OT: Number of observed tests
  - OP: Number of observed positive tests
  - For a closed population with size N, if the testing follows the bottom-up model,
    - Assume $OT(t) \sim Binom(N,\ T(Y(t); T_B, T_Y)$
    - Assume $OP(t) \sim Binom\big(OT, P(Y(t); T_B, T_Y)\big)$
  - Follow the bottom-up model, one could fit observed data using MLE, with the likelihood

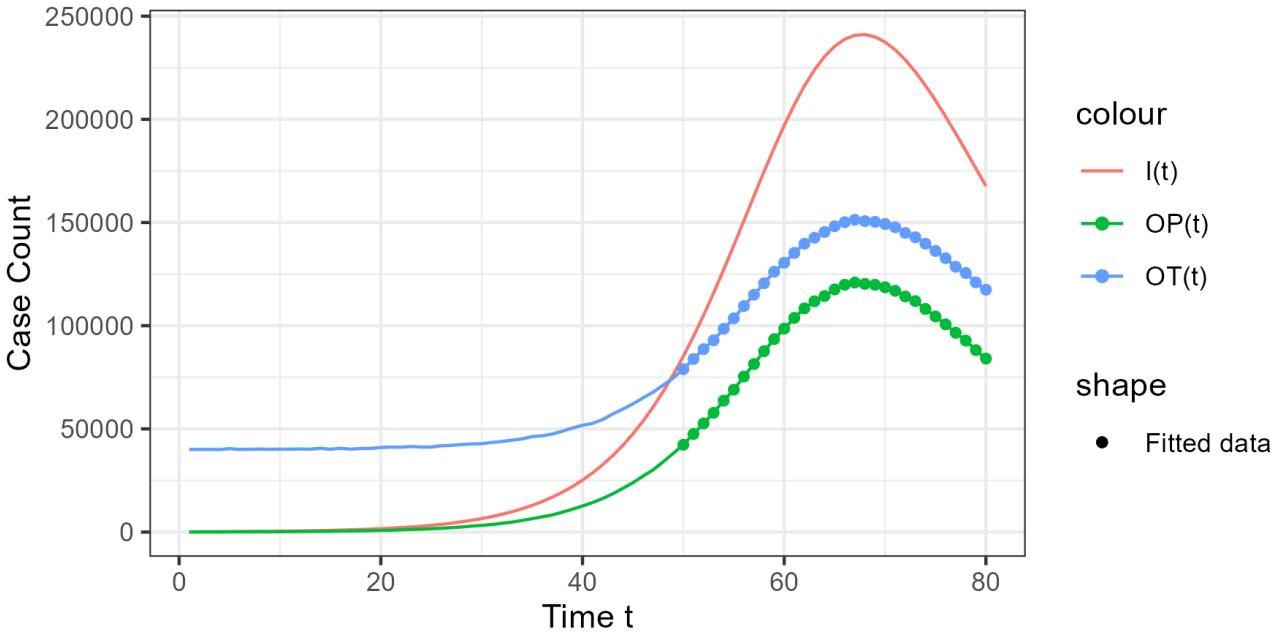
$$L = \prod_{\mathrm{t}} f_{\mathrm{B}}\big(OT(\mathrm{t}); N, T(\hat{Y}(t);\ \hat{T}_B, \hat{T}_{\mathrm{Y}})\big) \times f_{\mathrm{B}}\big(OP(\mathrm{t}); \mathrm{OT}(\mathrm{t}), T(\hat{Y}(t);\ \hat{T}_B, \hat{T}_{\mathrm{Y}})\big)$$

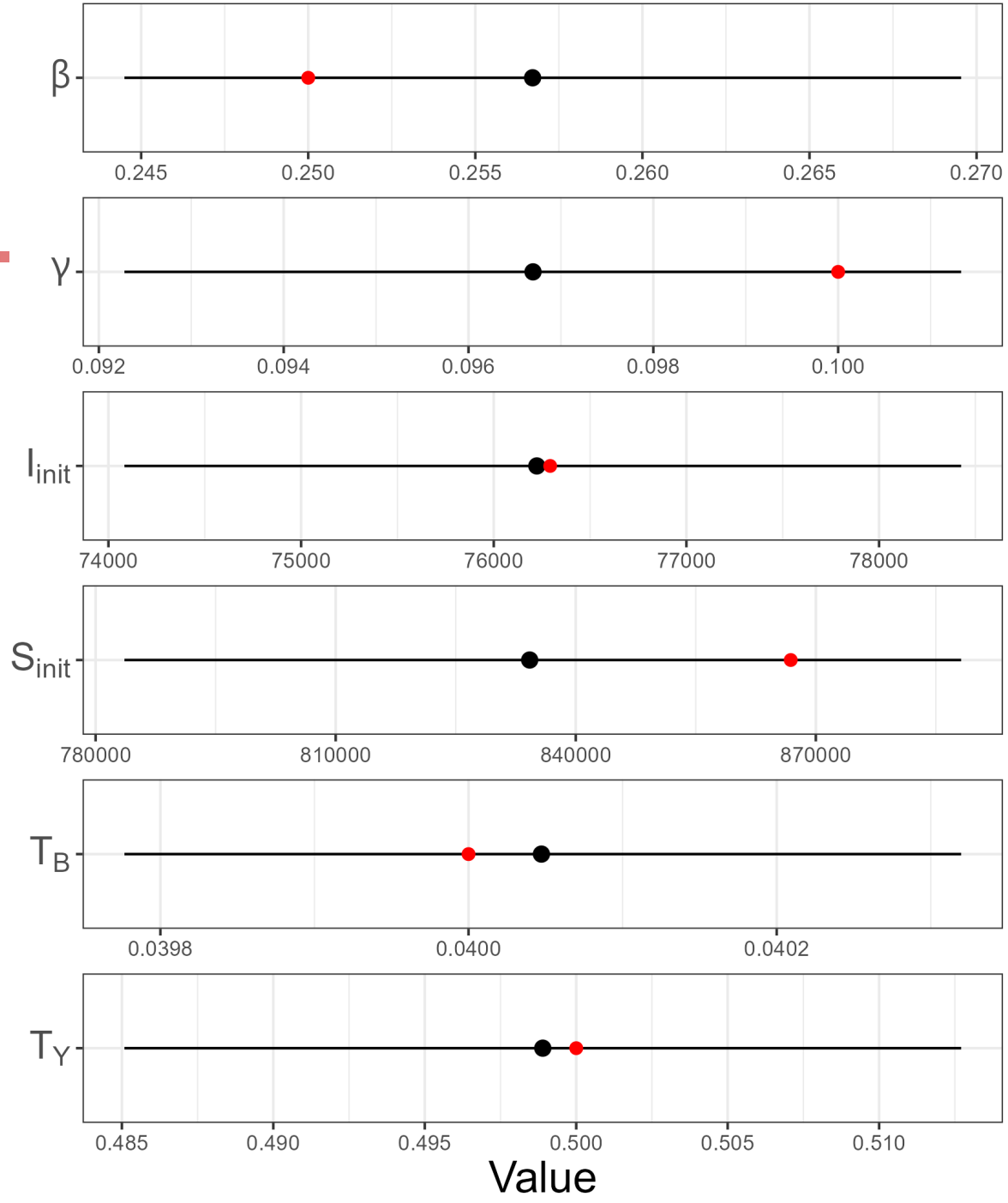
    - $f_B(x; N, p)$ is the PDF of Binomial distribution with parameter N, p
    - Fitting for $\hat{T}_B$, $\hat{T}_{\mathrm{Y}}$ and parameters for model of $\hat{Y}(t)$
  - Only need data of OT, OP and can be applied to general dynamic model for $Y(t)$

Basic SIR Model

- Population $N = 100{,}000$

$$\begin{cases} \dot{S} = -\beta I S/\mathrm{N} \\ I = \mathrm{N} - S - R \\ \dot{R} = \gamma I \end{cases}$$



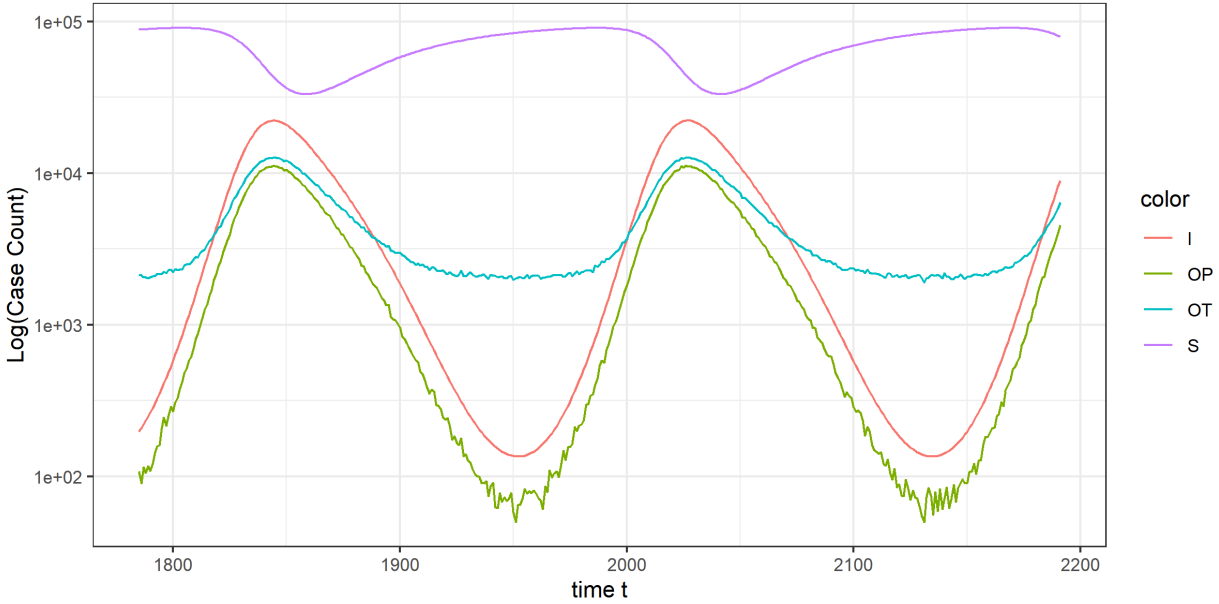


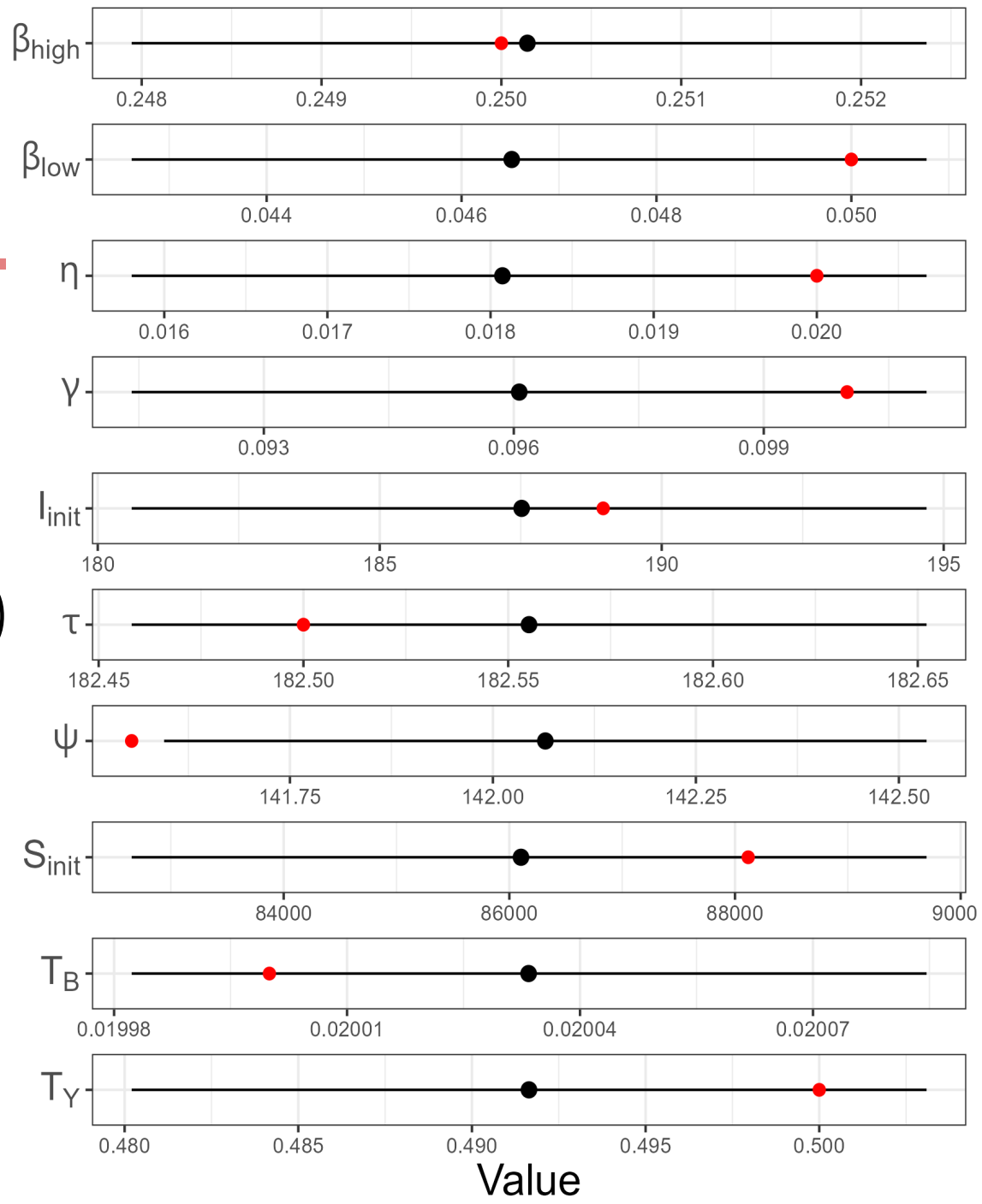
Seasonal SIRS Model

- Population $N = 100{,}000$

$$\begin{cases} \dot{S} = -\beta(\mathrm{t})IS/\mathrm{N} + \eta R \\ I = \mathrm{N} - S - R \\ \dot{R} = \gamma I - \eta R \end{cases}$$

$$\beta(t) = \beta_{\mathrm{low}} + \frac{1}{2}\left(cos\left((t+\psi)\frac{2\pi}{\tau}\right) + 1\right) \times \left(\beta_{\mathrm{high}} - \beta_{\mathrm{low}}\right)$$





What's Next

- Ongoing Investigations
  - Parameters with prior distribution
  - Time-varying baseline hazard/odds ratio B
  - Validating the model with real data (e.g. RVDSS)
- Future Investigations
  - Consider more risk-factors and mechanism for testing process
  - Modeling impact of testing policies and changes
  - Consider heterogeneity in behavior, testing and infection

Acknowledgement

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  - MacTheoBio Lab
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  - *macpan2: Fast and Flexible Compartmental Modelling*. R package version 3.5.1, https://canmod.github.io/macpan2/
  - Dr. Michael Li & Dr. David Champredon (PHAC)
- ariData Collaboration Project
  - Acute Respiratory Infections: RSV, Influenza, COVID-19 etc.
  - Bridging expertise within academia, data administrators and public health agencies at different level in Canada
  - Using similar modelling approaches to combine multiple data streams
  - Improving surveillance data collection, sharing, analysis, modelling and forecasting
  - Further enlighten decision making, planning, preparedness and intervention

macpan2